AI Fundamentals Cheat Sheet

Core Concepts · Architectures · LLMs · Agents
Metrics · Prompting · Safety & Ethics

June 2026 · For practitioners, engineers & enthusiasts

1 · Core AI Concepts

Term	Definition
Artificial Intelligence (AI)	Simulation of human intelligence by machines — reasoning, learning, perception, problem-solving.
Machine Learning (ML)	AI subfield where systems learn from data without being explicitly programmed.
Deep Learning (DL)	ML using multi-layered neural networks to learn hierarchical data representations.
Foundation Model	Large model trained on broad data (text, images, code) adaptable to many downstream tasks.
Generative AI	AI that creates new content — text, images, audio, video, code — from learned patterns.
Inference	Using a trained model to generate predictions or outputs on new, unseen data.
Training	Adjusting model parameters by minimizing a loss function over a dataset.
Fine-tuning	Further training a pre-trained model on domain-specific data to specialise it.
Embeddings	Dense numerical vector representations of data (words, images) in a latent space.
Latent Space	The abstract multi-dimensional space where a model stores compressed representations.
Token	Basic unit of text for LLMs — roughly ¾ of a word. "unhappy" ≈ 2 tokens.
Context Window	Maximum number of tokens a model can process in a single forward pass.
Parameters	Learnable weights in a neural network — model knowledge encoded in billions of these.
Prompt	Input text/instruction given to an AI model to elicit a desired response.

4 · Large Language Models (LLMs)

2 · Machine Learning Types

Type	How It Works	Examples
Supervised	Learns from labeled input-output pairs — ground truth provided for every sample.	Classification, Regression; Linear reg, SVMs, Neural nets
Unsupervised	Finds hidden patterns in unlabeled data — no explicit ground truth.	Clustering, Dim. reduction; k-Means, PCA, Autoencoders
Reinforcement (RL)	Agent takes actions in an environment to maximise cumulative reward.	Game playing, Robotics, RLHF; PPO, DQN, SAC
Semi-Supervised	Combines a small labeled set with a large unlabeled set.	Label propagation, Self-training, Graph-based methods
Self-Supervised	Model generates its own training signal from raw data (masked tokens).	LLM pre-training, BERT, GPT, SimCLR

Type	How It Works	Examples
Transfer Learning	Reuses pre-trained weights to accelerate learning on a new task.	Fine-tuning GPT; ImageNet → medical imaging

3 · Neural Network Architectures

Arch.	Full Name	Best Used For
MLP	Multi-Layer Perceptron	Tabular data, basic classification
CNN	Convolutional Neural Network	Images, spatial data, pattern recognition
RNN	Recurrent Neural Network	Sequential/time-series data (legacy)
LSTM	Long Short-Term Memory	Long sequences, NLP (pre-Transformer era)
GRU	Gated Recurrent Unit	Simpler, faster alternative to LSTM
Transformer	Transformer	Sequence modeling, NLP, multimodal AI — dominant today
GAN	Generative Adversarial Network	Image/data synthesis and generation
VAE	Variational Autoencoder	Generative modeling, latent space learning
Diffusion	Diffusion Model	High-quality image/audio/video generation
GNN	Graph Neural Network	Graph-structured data, knowledge graphs
MoE	Mixture of Experts	Efficient large-scale models (e.g. Mixtral, GPT-4)
SSM	State Space Model	Efficient long-context modeling (e.g. Mamba)

Transformer — Key Components

Component	Role
Tokenisation + Embedding	Convert text tokens to dense vectors
Positional Encoding	Inject sequence order information into embeddings
Multi-Head Self-Attention	Learn relationships between all token pairs via Q, K, V matrices
Feed-Forward Network (FFN)	Apply non-linear transformation to each position independently
Layer Norm + Residual	Stabilise training; allow gradients to flow through deep stacks
Softmax Output Head	Convert final hidden states to probability distribution over vocab

4 · Large Language Models (LLMs)

Term	Definition
Pre-training	Training on massive text corpus (trillions of tokens) via next-token prediction.
RLHF	Reinforcement Learning from Human Feedback — aligns outputs to human preferences via a reward model.

Term	Definition
RLAIF	RL from AI Feedback — uses an AI model as preference labeler instead of humans.
SFT	Supervised Fine-Tuning — instruction-following training on curated input-output pairs.
DPO	Direct Preference Optimisation — trains on preferred vs rejected response pairs directly.
RAG	Retrieval-Augmented Generation — augments prompts with chunks from an external knowledge base.
Temperature	Randomness dial: 0 = deterministic/greedy. Higher values = more varied, creative output.
Top-p (Nucleus)	Sample from smallest token set whose cumulative probability ≥ p.
Hallucination	Model generates confident but factually incorrect or fabricated information.
Grounding	Anchoring outputs to verifiable external facts or documents to reduce hallucination.
KV Cache	Caches Key-Value attention states to substantially speed up autoregressive decoding.
System Prompt	Hidden instructions before user input that set model role, tone, and safety behaviour.

Model Size Reference

Scale	Params	Characteristics
Small (SLM)	1–7B	Runs on-device/CPU; fast; limited multi-step reasoning
Medium	13–34B	Good performance/cost ratio; strong at most everyday tasks
Large	70–90B	Strong reasoning; requires server-grade GPU (A100/H100)
Frontier	100B–1T+	SOTA; multi-modal; agentic; e.g. GPT-4o, Claude 3.5, Gemini Ultra

5 · Agentic AI

Term	Definition
AI Agent	Autonomous system that perceives inputs, reasons, plans, and takes actions to pursue multi-step goals.
ReAct	Reason + Act — interleaves chain-of-thought reasoning with tool calls in a loop until goal is achieved.
Chain-of-Thought (CoT)	Prompting that elicits explicit step-by-step reasoning before answering.
Tool Use / Function Calling	Agent invokes external APIs, databases, code interpreters, or other tools mid-task.
Multi-Agent System	Multiple AI agents collaborating, each with distinct roles, skills, or knowledge domains.
Orchestrator	Agent that coordinates sub-agents and manages task decomposition and workflow execution.
Memory (Agentic)	Short-term: context window. Long-term: vector DB / episodic logs / key-value stores.
MCP	Model Context Protocol — open standard connecting agents to tools, resources, and data sources.
ACP	Agent Communication Protocol — standard for structured inter-agent messaging and coordination.

Term	Definition
A2A	Agent-to-Agent (Google) — protocol for cross-agent interoperability and task delegation.
Prompt Injection	Attack that embeds malicious instructions in external data to hijack agent behaviour.

Agentic Memory Types

Type	Where Stored	Description
In-Context	Context window	Working memory — conversation, scratchpad. Lost at context limit.
External	Vector DB / KV	Persistent retrieval-based memory across sessions and agent runs.
Episodic	Log / Database	Records of past agent experiences, task outcomes, and actions.
Semantic	Knowledge graph	Factual/conceptual knowledge about the world or problem domain.
Procedural	Prompts / Code	How-to knowledge — encoded in system prompts, tools, or functions.

6 · Key Metrics & Evaluation

Classification Metrics

Metric	Formula / Description	Best When
Accuracy	(TP+TN) / Total	Classes are balanced — not reliable for imbalanced data
Precision	TP / (TP+FP)	False positives are costly (e.g. spam filter, fraud detection)
Recall	TP / (TP+FN)	False negatives are costly (e.g. cancer screening, safety systems)
F1 Score	2×(P×R)/(P+R)	Need balance between precision and recall
AUC-ROC	Area under ROC curve	Threshold-independent evaluation of classifier ranking quality
MCC	Matthews Correlation Coefficient	Imbalanced classes — more informative than F1

Regression Metrics

Metric	Description
MAE	Mean Absolute Error — average |prediction − actual|; robust to outliers
MSE	Mean Squared Error — penalises large errors heavily; units² of target
RMSE	Root MSE — same unit as target; easier to interpret than MSE
R^2	Proportion of variance explained (0–1); 1.0 = perfect model, 0 = baseline mean

LLM Evaluation Benchmarks

Metric / Benchmark	Description
Perplexity	How well model predicts test data — lower = better. Measures language model quality.
BLEU	N-gram overlap for machine translation (Bilingual Evaluation Understudy)
ROUGE	Recall-oriented overlap for summarisation; ROUGE-1, ROUGE-2, ROUGE-L variants

Metric / Benchmark	Description
BERTScore	Semantic similarity using BERT embeddings — better than surface n-gram overlap
MMLU	Massive Multitask Language Understanding — 57 academic subject areas benchmark
HumanEval	Code generation benchmark — pass@k on 164 Python programming tasks (OpenAI)
MT-Bench	Multi-turn conversation quality evaluated by GPT-4 as judge
TruthfulQA	Measures tendency to give truthful vs imitative-false (common misconception) answers

7 · Training Concepts

Term	Definition
Epoch	One complete pass through the entire training dataset.
Batch Size	Number of samples processed before model weights are updated once.
Learning Rate	Gradient descent step size — too high = unstable; too low = slow convergence.
Backpropagation	Computing gradients of loss w.r.t. each weight via chain rule through layers.
Overfitting	Model memorises training data; performs poorly on unseen examples (high variance).
Underfitting	Model too simple to capture data patterns; high bias, poor training performance.
Regularisation	Techniques to reduce overfitting: L1/L2 weight penalty, dropout, early stopping.
Transfer Learning	Reusing pre-trained weights as starting point — dramatically reduces training cost.
PEFT	Parameter-Efficient Fine-Tuning — update a tiny fraction of weights; rest frozen.
LoRA	Low-Rank Adaptation — inject trainable low-rank matrices into frozen model layers.
QLoRA	Quantized LoRA — fine-tune a quantised (INT4) model using LoRA adapters.
Quantisation	Reduce weight precision (FP32→INT8/INT4) to cut memory and inference compute.
Pruning	Remove redundant weights/neurons to reduce model size and speed up inference.
Distillation	Train small "student" model to mimic outputs of a large "teacher" model.
Warm-up	Gradually increase learning rate from near-zero at training start for stability.

8 · Prompt Engineering

Technique	Description	Signal / Example
Zero-shot	No examples given — model uses only task description.	Classify this sentiment: …
Few-shot	2–5 labeled examples included in the prompt.	Add sample Q→A pairs before the query
Chain-of-Thought	Elicit step-by-step reasoning before the final answer.	Let's think step by step…
Self-Consistency	Sample multiple CoT paths; take the majority answer.	Run same prompt 5× then vote

Technique	Description	Signal / Example
Tree of Thoughts	Explore multiple reasoning branches; evaluate and prune.	Complex multi-step problem solving
ReAct	Interleave reasoning traces with tool calls in a loop.	Agents: Thought → Act → Observe
Role Prompting	Assign an expert persona to steer tone and knowledge.	You are a senior data scientist…
Constrained Output	Specify exact output format for parseable responses.	Return as JSON: {'label': …}
Least-to-Most	Break complex problem into sub-problems, solve in order.	Multi-step maths, coding tasks

Prompt Anatomy

Component	Purpose	Example
System / Role	Set persona, tone, and constraints	"You are a helpful, concise assistant"
Context / Background	Provide relevant facts or documents	Paste relevant article or document excerpt
Task / Instruction	Clearly state what to do	"Summarise the following in 3 bullets:"
Format	Specify output structure	"Return as markdown table"
Examples	Demonstrate expected output	1–3 input→output demonstrations

9 · AI Safety & Ethics

Term	Definition
Alignment	Ensuring AI goals, values, and behaviours match human intentions and preferences.
RLHF	Key alignment technique — trains a reward model from human preferences, then fine-tunes via RL.
Constitutional AI	Anthropic's method: model critiques and revises outputs against written principles.
Guardrails	Filters, classifiers, and rules preventing harmful or off-policy model outputs.
Jailbreaking	Adversarial prompts designed to bypass model safety restrictions and guardrails.
Prompt Injection	Malicious input embedded in data that hijacks agent instructions or system prompt.
Bias	Systematic skew in model outputs due to imbalanced or unrepresentative training data.
Explainability (XAI)	Methods to interpret model decisions — SHAP, LIME, attention visualisation, probing.
Data Privacy	Ensuring PII is not memorised, leaked, or reconstructable from model outputs.
Model Card	Documentation of model purpose, capabilities, limitations, and intended use.
Responsible AI (RAI)	Framework ensuring AI is fair, accountable, transparent, robust, privacy-preserving.
Dual-Use Risk	Models capable of both beneficial and harmful applications require careful governance.

Term	Definition
EU AI Act	EU regulation classifying AI by risk level (unacceptable, high, limited, minimal).

8 · Prompt Engineering

Technique	Description	Example Signal
Zero-shot	No examples — model uses task description alone.	"Classify this sentiment: …"
Few-shot	2–5 labeled examples included in the prompt.	Include sample Q→A pairs
Chain-of-Thought	Elicit step-by-step reasoning before the final answer.	"Think step by step…"
Self-Consistency	Sample multiple CoT paths; take the majority answer.	Run same prompt 5× then vote
Tree of Thoughts	Explore multiple reasoning branches; evaluate & prune.	Complex multi-step problems
ReAct	Interleave reasoning traces with tool calls in a loop.	Thought → Act → Observe
Role Prompting	Assign an expert persona to steer tone and knowledge.	"You are a senior ML engineer…"
Constrained Out.	Specify exact output format for parseable responses.	"Return as JSON: {'label': …}"
Least-to-Most	Decompose problem into sub-problems, solve in order.	Multi-step maths, coding tasks
System Prompt	Hidden instructions setting model role and boundaries.	Used in all production deployments

Prompt Anatomy

Component	Purpose	Example
System / Role	Set persona, tone, and constraints	"You are a helpful, concise assistant…"
Context	Provide relevant background facts or documents	Paste a relevant article or data
Task / Instruction	Clearly state what to do	"Summarise in 3 bullet points:"
Format	Specify output structure	"Return as a markdown table"
Examples	Demonstrate expected output style	1–3 labeled input → output pairs

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Responsible AI (RAI)	Framework ensuring AI is fair, accountable, transparent, robust, and privacy-preserving.
Dual-Use Risk	Models capable of both beneficial and harmful applications (e.g. drug design / bioweapons).
EU AI Act	EU regulation classifying AI by risk level with proportionate legal obligations per tier.

10 · Acronyms Reference

Acronym	Full Form	Acronym	Full Form
A2A	Agent-to-Agent (protocol)	**MoE**	Mixture of Experts
ACP	Agent Communication Protocol	**MSE**	Mean Squared Error
AGI	Artificial General Intelligence	**NLG**	Natural Language Generation
AI	Artificial Intelligence	**NLP**	Natural Language Processing
ANI	Artificial Narrow Intelligence	**NLU**	Natural Language Understanding
API	Application Programming Interface	**NLI**	Natural Language Inference
ASI	Artificial Super Intelligence	**OCR**	Optical Character Recognition
AUC	Area Under the Curve	**OOD**	Out-of-Distribution
BM25	Best Match 25 (keyword search)	**PCA**	Principal Component Analysis
BLEU	Bilingual Evaluation Understudy	**PEFT**	Parameter-Efficient Fine-Tuning
CoT	Chain of Thought	**PPO**	Proximal Policy Optimization
CV	Computer Vision	**QA**	Question Answering
DL	Deep Learning	**QLoRA**	Quantized Low-Rank Adaptation
DPO	Direct Preference Optimization	**RAG**	Retrieval-Augmented Generation
FAISS	Facebook AI Similarity Search	**RAI**	Responsible AI
FFN	Feed-Forward Network	**ReAct**	Reason + Act (agent pattern)
FN	False Negative	**RL**	Reinforcement Learning
FP	False Positive	**RLAIF**	RL from AI Feedback
FP16	16-bit Floating Point (half precision)	**RLHF**	RL from Human Feedback
FP32	32-bit Floating Point (full precision)	**RMSE**	Root Mean Squared Error

Acronym	Full Form	Acronym	Full Form
FSL	Few-Shot Learning	**RNN**	Recurrent Neural Network
GAN	Generative Adversarial Network	**ROC**	Receiver Operating Characteristic
GPT	Generative Pre-trained Transformer	**ROUGE**	Recall-Oriented Understudy for Gisting Evaluation
GNN	Graph Neural Network	**SFT**	Supervised Fine-Tuning
GRU	Gated Recurrent Unit	**SHAP**	SHapley Additive exPlanations
HPC	High-Performance Computing	**SLM**	Small Language Model
ICL	In-Context Learning	**SSM**	State Space Model
KV Cache	Key-Value Cache (attention)	**SVM**	Support Vector Machine
LLM	Large Language Model	**TN**	True Negative
LoRA	Low-Rank Adaptation	**ToT**	Tree of Thoughts
LSTM	Long Short-Term Memory	**TP**	True Positive
MAE	Mean Absolute Error	**UMAP**	Uniform Manifold Approximation and Projection
MCP	Model Context Protocol	**VAE**	Variational Autoencoder
ML	Machine Learning	**VLM**	Vision-Language Model
MLP	Multi-Layer Perceptron	**XAI**	Explainable AI
MMLU	Massive Multitask Language Understanding	**ZSL**	Zero-Shot Learning

10 · Vector Databases

A **vector database** stores high-dimensional embeddings and enables fast similarity search — the backbone of RAG, semantic search, recommendation engines, and long-term memory for AI agents. Text, images, or audio are first converted to embedding vectors by a model, then stored and indexed for retrieval.

Core Concepts

Concept	Definition
Vector Embedding	A list of numbers (e.g. 1,536 floats) that encodes the meaning of text, image, or audio.
Similarity Search	Find embeddings geometrically closest to a query vector — returns semantically similar content.
Cosine Similarity	Measures angle between vectors (–1 to 1); most common metric for text embeddings.
Dot Product	Measures projection of one vector onto another; used when vectors are L2-normalised.
ANN	Approximate Nearest Neighbour — fast (not exact) similarity search; enables scale.
HNSW	Hierarchical Navigable Small World — most popular ANN index; fast, accurate, and widely used.
IVF	Inverted File Index — clusters vectors into buckets; faster for very large (100M+) datasets.
Namespace / Collection	Logical partition in a vector DB to isolate different datasets, users, or tenants.
Metadata Filtering	Filter search results by attached metadata fields (e.g. date, source, category, user_id).

Concept	Definition
Hybrid Search	Combine dense (vector) and sparse (BM25 keyword) retrieval for improved recall.
Re-ranking	Post-retrieval step re-scoring candidates with a cross-encoder for higher precision.
Chunking	Splitting documents into smaller pieces (chunks) before embedding for better retrieval granularity.

Popular Vector Databases

Database	Type	Key Characteristics
Pinecone	Managed SaaS	Fully managed; serverless tier; very easy to get started
Weaviate	Open-source / Cloud	Multi-modal; built-in hybrid search; rich query API
Qdrant	Open-source / Cloud	Rust-based; very fast; rich payload filtering capabilities
Chroma	Open-source	Local-first; Python-native; popular for rapid prototyping
FAISS (Meta)	Library	Efficient CPU/GPU index; offline use; not a DB — no persistence
pgvector	Postgres extension	Vector search inside existing Postgres; great for small–medium scale
Amazon OpenSearch	AWS Managed	OpenSearch k-NN plugin; integrates tightly with AWS ecosystem
Amazon Aurora pgvector	AWS Managed	Postgres-compatible; vectors + relational data in one database

11 · Multimodal AI

Multimodal AI processes and generates multiple types of data — text, images, audio, video, and structured data — within a single unified model. Modern frontier models like GPT-4o, Gemini Ultra, and Claude 3.5 Sonnet are all multimodal, enabling richer real-world understanding and generation.

Key Concepts

Term	Definition
Vision-Language Model (VLM)	Model that processes both images and text — can describe images, answer visual questions, extract info from charts.
Image Encoder	Converts an image into a sequence of patch embeddings (e.g. ViT — Vision Transformer).
Cross-Attention	Mechanism allowing one modality (e.g. text) to attend to another (e.g. image patches).
CLIP	Contrastive Language-Image Pre-training (OpenAI) — learns shared text-image embedding space.
Stable Diffusion	Text-to-image diffusion model; open-source; widely used for image generation tasks.
DALL·E / Imagen	Proprietary text-to-image models from OpenAI and Google respectively.
Whisper	OpenAI speech recognition model — transcribes audio to text in 99 languages.
Text-to-Speech (TTS)	AI models converting text to natural-sounding spoken audio (e.g. ElevenLabs, OpenAI TTS).
Video Understanding	Models that process video frames + audio to answer questions or generate captions.

Term	Definition
Document AI	Models extracting structured information from documents: PDFs, forms, invoices (e.g. OCR + LLM).
Multi-modal RAG	RAG systems that retrieve and use images, tables, or charts alongside text passages.
Grounded Generation	Model references specific regions of an image or document when generating its response.

Modality Types

Modality	Input Examples	Output Examples
Text	Prompts, documents, code	Summaries, answers, code generation
Image	Photos, diagrams, screenshots	Captions, edited images, new images
Audio	Speech, music, sound effects	Transcriptions, voice responses, music
Video	Clips, screen recordings	Video captions, edited clips, analysis
Structured Data	Tables, JSON, spreadsheets	Analysis, charts, NL summaries

12 · AWS AI & ML Services

AWS provides a layered AI stack: **Infrastructure** (raw compute/chips), **Platform** (model training and deployment tools), and **Applications** (pre-built AI capabilities via APIs). Most practitioners interact with the middle and top layers.

Foundation Models & Generative AI

Service	What It Does
Amazon Bedrock	Fully managed access to leading FMs (Anthropic Claude, Meta Llama, Mistral, Amazon Nova, Stability AI, Cohere) via a single API. Supports RAG, agents, fine-tuning, and guardrails.
Amazon Nova	Amazon's own family of foundation models (Nova Micro, Lite, Pro, Premier) — text, image, video understanding and generation.
Bedrock Agents	Build fully managed AI agents with tool use, knowledge bases (RAG), memory, and guardrails on Bedrock.
Bedrock Knowledge Bases	Managed RAG — ingest documents into a vector store; query with automatic retrieval and augmentation.
Bedrock Guardrails	Content filtering, PII redaction, topic blocking, and grounding checks for Bedrock models.
Amazon Q	Enterprise AI assistant powered by Bedrock; connects to company data, Jira, Salesforce, etc.
PartyRock	No-code playground to prototype Bedrock-powered generative AI apps — great for beginners.

ML Platform (Build & Train)

Service	What It Does
Amazon SageMaker	End-to-end ML platform: notebooks, training, hyperparameter tuning, model registry, deployment, and MLOps pipelines.

Service	What It Does
SageMaker Studio	Web-based IDE for data science and ML — notebooks, experiments, feature store, and model monitoring.
SageMaker Autopilot	AutoML — automatically builds, trains, and tunes models from tabular data with full explainability.
SageMaker JumpStart	Model hub with 300+ pre-trained models (including FMs) ready to deploy with one click.
SageMaker Pipelines	CI/CD pipelines for ML — automate training, evaluation, and deployment workflows.
AWS Trainium	Custom AWS chip (Trn1/Trn2 instances) optimised for training large deep learning models.
AWS Inferentia	Custom AWS chip (Inf2 instances) for high-throughput, low-cost model inference.

Pre-Built AI Services (No ML Expertise Needed)

Service	Capability
Amazon Rekognition	Computer vision: object detection, facial recognition, content moderation, text-in-image.
Amazon Textract	Document AI: extract text, tables, forms, and key-value pairs from PDFs and images.
Amazon Comprehend	NLP: sentiment analysis, entity recognition, topic modelling, PII detection.
Amazon Transcribe	Speech-to-text: real-time and batch audio transcription with speaker identification.
Amazon Polly	Text-to-speech: lifelike speech in 60+ voices and 30+ languages (TTS).
Amazon Translate	Neural machine translation across 75+ languages.
Amazon Forecast	Managed time-series forecasting using AutoML (demand planning, capacity forecasting).
Amazon Personalize	Real-time personalisation and recommendation engine (similar to Amazon.com's engine).
Amazon Kendra	Intelligent enterprise search powered by ML — search across documents and data silos.

13 · AI/ML Workflow & MLOps

Building AI in production requires more than training a model. **MLOps** (ML Operations) applies DevOps principles to ML — automating the full lifecycle from data prep to model deployment and monitoring.

End-to-End ML Lifecycle

Stage	Activities	Key Tools / Concepts
1. Problem Definition	Define business objective, success metric, and ML problem type.	KPIs, baseline metrics, feasibility
2. Data Collection	Gather raw data from databases, APIs, files, sensors, web scraping.	Data pipelines, ETL, lakes
3. Data Preparation	Clean, label, split (train/val/test), engineer features.	Pandas, Spark, Feature Store
4. Model Selection	Choose algorithm/architecture; baseline with simple models first.	AutoML, model zoos, SageMaker JumpStart

Stage	Activities	Key Tools / Concepts
5. Training	Fit model on training data; tune hyperparameters; track experiments.	SageMaker, MLflow, Weights & Biases
6. Evaluation	Measure performance on held-out test set with appropriate metrics.	F1, AUC, RMSE, LLM benchmarks
7. Deployment	Serve model via API endpoint; A/B testing; canary rollouts.	SageMaker Endpoints, Lambda, ECS
8. Monitoring	Track data drift, model drift, latency, errors in production.	SageMaker Clarify/Monitor, CloudWatch
9. Iteration	Retrain with new data; update model; continuous improvement loop.	CI/CD for ML, automated retraining

Key MLOps Concepts

Term	Definition
Data Drift	Distribution of input data changes over time, degrading model performance.
Model Drift	Model predictions become less accurate as the real-world environment evolves.
Feature Store	Centralised repository for ML features — enables reuse, consistency, and low-latency serving.
Model Registry	Versioned catalogue of trained models with metadata, lineage, and deployment status.
CI/CD for ML	Automated pipelines for testing, training, evaluating, and deploying ML models.
Shadow Mode	New model runs in parallel receiving real traffic but its outputs are not shown to users.
A/B Testing	Route a percentage of live traffic to a challenger model to compare performance.
Explainability	Techniques to understand why a model made a specific prediction (SHAP, LIME, integrated gradients).

15 · AWS Data Foundations for AI

Data is the foundation that makes AI possible. **AWS Data Foundations for AI** is AWS's strategic framework for building the critical data infrastructure organisations need — beyond just storing data, covering processing, governance, and accessibility.

The Three Strategic Pillars

Pillar	What It Means	Why It Matters
1. Open Data Architecture	Use data where it lives through open standards — no vendor lock-in, no forced data migration.	Work with data across any system without being tied to one vendor's tools.
2. Foundational Excellence	Enterprise-grade reliability and scalability that handles unpredictable AI workloads automatically.	Removes the need for constant manual infrastructure management as AI demand surges.
3. Trusted Data for AI	Built-in governance, security, and data quality controls ensuring AI works with reliable information.	AI is only as good as its data — bad or ungoverned data leads to inaccurate or risky outputs.

The Four Architectural Layers

Layer	Name	What Happens Here
Layer 1	Data Producers & Consumers	Interfaces for people, apps, and AI agents to create and consume data.
Layer 2	Catalog, Semantics & Memory	Helps systems understand what data means, where it lives, and how datasets relate.
Layer 3	Data Processing	Transforms raw data into usable information — cleaning, deduplicating, standardising formats.
Layer 4	Data Storage	Manages where data physically lives (databases, lakes, warehouses) with automatic capacity management.

Data Types & Storage

Concept	Definition
Structured Data	Organised in rows and columns like a spreadsheet — each value has a defined place.
Unstructured Data	Doesn't fit neatly into tables — documents, images, video, audio, social media.
Database	Stores structured information for real-time application use (accounts, inventory).
Data Lake	Stores large amounts of raw data in its original format until needed for analysis.
Data Warehouse	Centralised store of cleaned, organised data from multiple sources — built for analysis.
Data Producers	People, apps, and systems that create data (business apps, sensors, employees).
Data Consumers	People, apps, and AI agents that use data to gain insights or make decisions.
Scattered Data	Data spread across many different systems — the #1 obstacle to effective AI.

16 · Building Responsible AI

Responsible AI means building systems that are fair, trustworthy, and safe throughout their lifecycle. AWS defines responsible AI through **eight core dimensions**.

The Eight Dimensions of Responsible AI

Dimension	What It Means	Example
Fairness	AI treats all user groups equitably — no unjust discrimination.	Loan model performing equally across all demographic groups.
Explainability	Humans can understand why the AI made a specific decision.	Fraud detection showing which signals triggered an alert.
Privacy & Security	Sensitive data and models are protected from leakage and attacks.	PII is never stored in model weights or returned in outputs.
Safety	AI prevents harmful outputs and cannot be misused to cause damage.	Content filters blocking toxic or dangerous responses.
Controllability	Humans can monitor, adjust, override, or shut down AI at any time.	Guardrails restricting topics; kill switches for rogue agents.
Veracity & Robustness	Maintains accuracy even with unexpected or adversarial inputs.	Graceful degradation on noisy or out-of-distribution data.

Dimension	What It Means	Example
Governance	Policies and accountability structures guide AI development and use.	Defined prohibited use cases; legal review for high-risk AI.
Transparency	AI capabilities and limitations are clearly communicated to users.	Model cards documenting training data, biases, and intended use.

Key Challenges & Safety Features

Challenge / Feature	Definition
Hallucinations	AI generates convincing but factually false information — a core reliability risk for all LLMs.
Toxicity	Output that is harmful, threatening, insulting, or demeaning to individuals or groups.
IP & Copyright	Risk of AI reproducing copyrighted content or artistic styles without permission.
Prohibited Use Cases	Applications where AI use is entirely banned — violations can result in legal fines.
High-Risk Use Cases	Not prohibited but require legal review due to elevated potential for harm.
Bedrock Guardrails	Content Filters (88% accuracy), Automated Reasoning (99% accuracy), Contextual Grounding.
Model Drift	Model becomes less accurate as real-world data shifts away from the training distribution.
Watermark	Invisible identifier embedded in AI-generated content for tracking and attribution.

17 · Amazon Bedrock — Deep Dive

Amazon Bedrock is a fully managed service that provides access to over 100 foundation models from leading AI companies (Anthropic, Meta, Mistral, Stability AI, Cohere, and Amazon) through a single API — no infrastructure management required.

Five Main Use Cases

Use Case	Description	Example
Text Generation	Generate, summarise, translate, and classify text.	Contract summarisation, email drafting
Advanced Search	Semantic search across documents using vector embeddings.	Enterprise knowledge base search
Virtual Assistants	Build conversational AI agents with memory and tools.	Customer service bot, sales assistant
Process Automation	Automate multi-step business workflows using agents.	Invoice processing, IT ticket resolution
Creative Design	Generate images, videos, and marketing content.	Product imagery, social media assets

Model Optimisation Techniques

Technique	What It Does	Benefit
Fine-tuning	Train an FM with your organisation's own data.	Better performance on domain-specific tasks

Technique	What It Does	Benefit
Knowledge Bases	Connect FMs to company documents via RAG.	Accurate, grounded, up-to-date answers
Model Distillation	Train a smaller model to mimic a larger one.	Lower cost, faster inference, same quality
Prompt Caching	Save commonly used instructions to avoid reprocessing.	Reduces costs by up to 90%
Intelligent Routing	Auto-select the best model for each query.	Saves up to 30% on inference costs
Model Evaluation	Test and compare models against real-world scenarios.	Choose the right FM for your use case

Amazon Bedrock AgentCore — 10 Integrated Tools

Tool	Function
Runtime	Runs agents automatically without managing servers — serverless agent execution.
Gateway	Connects agents to business tools and APIs with one-click integrations.
Policy	Controls and enforces rules for what agents can access and do.
Observability	Monitors agent performance, usage, costs, and behaviour in production.
Evaluations	Automatically monitors and checks agent performance and output quality.
Memory	Helps agents remember information across sessions and share context with other agents.
Identity	Controls who/what can access agent capabilities; maintains security boundaries.
Browser Tool	Lets agents interact with websites and web applications securely.
Code Interpreter	Allows agents to write, execute, and debug code safely in a sandbox.
AWS Agent Registry	Central catalogue to discover, manage, and govern AI agents and tools.

18 · Amazon Nova Model Family

Amazon Nova is AWS's family of cost-effective foundation models, available exclusively through Amazon Bedrock. Nova models are less expensive than other leading models while maintaining high performance across multiple content types.

Understanding Models (Text & Multimodal)

Model	Inputs	Best For
Nova Micro	Text only	Fast, simple tasks — quickest response times, lowest cost
Nova Lite	Text, images, video	Balancing speed and cost for everyday multimodal tasks
Nova 2 Lite	Text, images, video	Enhanced instruction-following and visual understanding
Nova Pro	Text, images, video	Best combination of accuracy, speed, and cost
Nova 2 Pro ■	Text, images, video	Enhanced reasoning for complex problem-solving tasks
Nova Premier	Text, images, video	Most powerful — large-scale analysis, complex tasks

Model	Inputs	Best For
Nova 2 Omni ■	Audio, image, video, text	Single model for all modalities; seamless cross-modal tasks

Creative & Speech Models

Model	Type	Capability
Nova Canvas	Image generation	Creates and edits professional images from text descriptions. Includes Inpainting (edit specific regions) and Outpainting (extend beyond borders). Watermarked.
Nova Reel	Video generation	Generates professional videos (incl. B-roll) from text and image inputs. Watermarked.
Nova Sonic	Speech-to-speech	Natural voice conversations with barge-in support; multiple language options.
Nova 2 Sonic	Speech-to-speech	Supports 7 languages; real-time streaming; natural conversation flow.
Nova MME	Multimodal embeddings	Converts text, images, video, audio into unified vectors — enables cross-modal semantic search.

Nova Extended Offerings

Service	What It Does
Amazon Nova Forge	Lets organisations build custom AI models on top of Nova foundations using their own data — maintaining security and responsible AI practices.
Amazon Nova Act	Agentic SDK for browser automation — helps developers build AI assistants that can perform web-based tasks autonomously.

■ = *In Preview at time of writing*

19 · Amazon Quick & Agentic AI Pathway

Amazon Quick is an AI-powered unified platform that helps business users find information, analyse data, and automate tasks — all in one secure workspace using natural language, without needing technical expertise.

Amazon Quick — Five Core Capabilities

Capability	What It Does	Best For
Quick Index	Securely brings together and organises all organisational information.	Making company knowledge searchable across all sources
Quick Sight	Turns complex data into visual insights and dashboards using natural language.	Business intelligence without BI expertise
Quick Research	Conducts comprehensive research combining internal and external sources.	Market analysis, competitive intelligence
Quick Flows	Creates automated workflows for routine tasks using everyday language.	Automating repeatable processes without coding

Capability	What It Does	Best For
Quick Automate	Manages complex business processes across departments using multiple agents.	End-to-end automation with human oversight

Agentic AI — The Four-Stage Pathway

Stage	What Happens	Key Activity
1. Perceive	Agent collects and processes data from various sources.	Observing environment, reading inputs, ingesting context
2. Reason	Interprets goals and develops an action plan using LLMs.	Thinking step-by-step, decomposing tasks, planning
3. Act	Executes tasks with monitoring and logging of all actions.	Calling tools, APIs, writing code, taking real actions
4. Learn	Improves through feedback from agents, LLMs, and humans.	Updating plans, correcting errors, building memory

AWS Agentic AI Solutions

Service / Tool	What It Does
Amazon Bedrock AgentCore	Deploys and operates AI agents at scale with 10 integrated tools (runtime, memory, security, etc.)
Strands Agents	Open-source SDK that simplifies the process of building AI agents in Python.
Amazon Connect	Family of agentic solutions: Connect Customer, Health, Decisions, and Talent.
Kiro	Agentic IDE — helps developers plan, write, and improve code through AI agent collaboration.
Kiro Autonomous Agent ■	Frontier coding agent that remembers developer preferences and handles multiple projects at once.
AWS Transform	Modernises enterprise systems by converting legacy software to modern cloud-ready applications.
AWS Security Agent	Automatically reviews and tests application security during development.
AWS DevOps Agent	Finds root causes of issues, coordinates team responses, prevents future incidents.

20 · AWS AI Services — At a Glance

	Amazon Bedrock	Amazon Nova	Amazon SageMaker AI	Amazon Quick
What It Is	Platform for building generative AI apps and agents using 100+ foundation models.	AWS's family of cost-effective FMs for text, images, video, and speech.	Fully managed service for building, training, and deploying custom AI models.	Unified workspace for business users to search data, research, and automate tasks.
Primary Use	Build and deploy AI apps/agents without managing infrastructure.	Process multiple content types with cost-effective multimodal models.	Build, train, customise, deploy, and manage AI models for specific business needs.	Interact with company knowledge and take action using natural language.

	Amazon Bedrock	Amazon Nova	Amazon SageMaker AI	Amazon Quick
Target Users	Developers and teams building production AI applications.	Teams needing versatile, affordable multimodal foundation models.	Data scientists, ML engineers, AI developers needing full lifecycle control.	Business users and analysts — no technical expertise required.
Main Benefit	Launch faster by eliminating infra management; enterprise security maintained.	Lower cost than other leading models; high performance across all modalities.	Complete control over model development with managed infrastructure support.	Research, analyse, and act using natural language without switching apps.

21 · Acronyms Reference

Acronym	Full Form	Acronym	Full Form
A2A	Agent-to-Agent Protocol	**AGI**	Artificial General Intelligence
AI	Artificial Intelligence	**ANI**	Artificial Narrow Intelligence
ANN	Approximate Nearest Neighbour	**API**	Application Programming Interface
ASI	Artificial Super Intelligence	**AUC**	Area Under the Curve
BLEU	Bilingual Evaluation Understudy	**BM25**	Best Match 25 (keyword search)
CoT	Chain of Thought	**CV**	Computer Vision
DL	Deep Learning	**DPO**	Direct Preference Optimisation
ETL	Extract, Transform, Load	**FAISS**	Facebook AI Similarity Search
FFN	Feed-Forward Network	**FM**	Foundation Model
FP	False Positive / Floating Point	**FP16**	16-bit Floating Point (half precision)
FP32	32-bit Floating Point (full precision)	**FSL**	Few-Shot Learning
GA	Generally Available	**GAN**	Generative Adversarial Network
GNN	Graph Neural Network	**GPT**	Generative Pre-trained Transformer
GPU	Graphics Processing Unit	**GRU**	Gated Recurrent Unit
HELM	Holistic Evaluation of Language Models	**HNSW**	Hierarchical Navigable Small World
HPC	High-Performance Computing	**ICL**	In-Context Learning
IDE	Integrated Development Environment	**IP**	Intellectual Property
IVF	Inverted File Index	**KV Cache**	Key-Value Cache
LoRA	Low-Rank Adaptation	**LSTM**	Long Short-Term Memory
MAE	Mean Absolute Error	**MCC**	Matthews Correlation Coefficient
MCP	Model Context Protocol	**ML**	Machine Learning
MLOps	Machine Learning Operations	**MLP**	Multi-Layer Perceptron
MMLU	Massive Multitask Language Understanding	**MoE**	Mixture of Experts
MSE	Mean Squared Error	**NLG**	Natural Language Generation
NLP	Natural Language Processing	**NLU**	Natural Language Understanding
OCR	Optical Character Recognition	**OOD**	Out-of-Distribution

Acronym	Full Form	Acronym	Full Form
PCA	Principal Component Analysis	**PEFT**	Parameter-Efficient Fine-Tuning
PPO	Proximal Policy Optimisation	**QA**	Question Answering
QLoRA	Quantized Low-Rank Adaptation	**RAG**	Retrieval-Augmented Generation
RAI	Responsible AI	**ReAct**	Reason + Act
RL	Reinforcement Learning	**RLAIF**	RL from AI Feedback
RLHF	RL from Human Feedback	**RMSE**	Root Mean Squared Error
RNN	Recurrent Neural Network	**ROC**	Receiver Operating Characteristic
ROUGE	Recall-Oriented Understudy for Gisting Evaluation	**SDK**	Software Development Kit
SFT	Supervised Fine-Tuning	**SHAP**	SHapley Additive exPlanations
SLM	Small Language Model	**SSM**	State Space Model
SVM	Support Vector Machine	**ToT**	Tree of Thoughts
TTS	Text-to-Speech	**VLM**	Vision-Language Model
XAI	Explainable AI	**ZSL**	Zero-Shot Learning

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